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Microsoft Windows [Version 10.0.26200.8875]
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```

```
C:\Users\jakka>cd C:\Program Files\MySQL\MySQL Server 8.0\bin
```

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 31
Server version: 8.0.46 MySQL Community Server - GPL
```

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```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> create database sqljoins;
Query OK, 1 row affected (0.05 sec)
```

```
mysql> use sqljoins;
Database changed
mysql> create table roles(role_id int primary key, role_name varchar(50));
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> create table users(user_id int primary key, name varchar(50), email varchar(50), role_id int);
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> create table departments(dept_id int primary key, dept_name varchar(50));
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> create table employees(emp_id int primary key, emp_name varchar(50), dept_id int, manager_id int, salary decimal(10,2));
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> create table customers(customer_id int primary key, customer_name varchar(50), city varchar(50));
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> create table orders(order_id int primary key, customer_id int, order_date date, total_amount decimal(10,2));
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> create table accounts(account_no int primary key, holder_name varchar(50), balance decimal(10,2));
Query OK, 0 rows affected (0.09 sec)
```

```
mysql> insert into roles values(1,'Admin'),(2,'Editor'),(3,'Viewer');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into users values(1,'Anil','anil@x.com',1),(2,'Priya','priya@x.com',2),(3,'Ravi','ravi@x.com',NULL);
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into departments values(1,'CSE'),(2,'ECE'),(3,'MECH');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into employees values(1,'Kiran',1,NULL,60000),(2,'Suresh',1,1,45000),(3,'Meena',2,NULL,55000),(4,'Arjun',2,3,40000);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> insert into customers values(1,'Rahul','Hyderabad'),(2,'Sneha','Chennai');
Query OK, 2 rows affected (0.10 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> insert into orders values(1,1,'2026-01-10',15000),(2,1,'2026-02-15',8000),(3,2,'2026-03-01',20000);
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into accounts values(101,'Ramesh',30000),(102,'Kavya',15000),(103,'Deepak',20000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> select emp_name,dept_name from employees e inner join departments d on e.dept_id=d.dept_id;
```

emp_name	dept_name
Kiran	CSE
Suresh	CSE
Meena	ECE
Arjun	ECE

```
4 rows in set (0.00 sec)
```

```
mysql> select name,email,role_name from users u left join roles r on u.role_id=r.role_id;
```

name	email	role_name
Anil	anil@x.com	Admin
Priya	priya@x.com	Editor
Ravi	ravi@x.com	NULL

```
3 rows in set (0.00 sec)
```

```
mysql> select name,role_name from users u right join roles r on u.role_id=r.role_id;
```

name	role_name
Anil	Admin
Priya	Editor
NULL	Viewer

```
mysql> select name,email,role_name from users u left join roles r on u.role_id=r.role_id union select name,email,role_name from users u right join roles r on u.role_id=r.role_id;
```

name	email	role_name
Anil	anil@x.com	Admin
Priya	priya@x.com	Editor
Ravi	ravi@x.com	NULL
NULL	NULL	Viewer

```
4 rows in set (0.00 sec)
```

```
mysql> select e.emp_name Employee, m.emp_name Manager from employees e left join employees m on e.manager_id=m.emp_id;
```

Employee	Manager
Kiran	NULL
Suresh	Kiran
Meena	NULL
Arjun	Meena

```
4 rows in set (0.00 sec)
```

```
mysql> select o.order_id,customer_name,city,total_amount from customers c join orders o on c.customer_id=o.customer_id;
```

order_id	customer_name	city	total_amount
1	Rahul	Hyderabad	15000.00
2	Rahul	Hyderabad	8000.00
3	Sneha	Chennai	20000.00

```
3 rows in set (0.00 sec)
```

```
mysql> select dept_name,count(*),avg(salary),max(salary) from employees e join departments d on e.dept_id=d.dept_id group by dept_name;
```

dept_name	count(*)	avg(salary)	max(salary)
CSE	2	52500.000000	60000.00
ECE	2	47500.000000	55000.00

```
2 rows in set (0.00 sec)
```

```
mysql> select emp_name,salary,dept_name from employees e join departments d on e.dept_id=d.dept_id where salary>50000 and dept_name='CSE';
```

emp_name	salary	dept_name
Kiran	60000.00	CSE

```
1 row in set (0.00 sec)
```

```
mysql> select customer_name,sum(total_amount) TotalPurchase from customers c join orders o on c.customer_id=o.customer_id group by customer_name order by TotalPurchase desc;
+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Rahul         | 23000.00      |
| Sneha         | 20000.00      |
+-----+-----+
2 rows in set (0.00 sec)

mysql> create table students(student_id int primary key, student_name varchar(50));
Query OK, 0 rows affected (0.08 sec)

mysql> create table faculty(faculty_id int primary key, faculty_name varchar(50));
Query OK, 0 rows affected (0.08 sec)

mysql> create table courses(course_id int primary key, course_name varchar(50), faculty_id int);
Query OK, 0 rows affected (0.08 sec)

mysql> create table enrollments(student_id int, course_id int);
Query OK, 0 rows affected (0.08 sec)

mysql> insert into students values(1,'Anu'),(2,'Bhanu');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> insert into faculty values(1,'Dr. Rao'),(2,'Dr. Iyer');
Query OK, 2 rows affected (0.05 sec)
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> insert into courses values(1,'DBMS',1),(2,'OS',2);
Query OK, 2 rows affected (0.04 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> insert into enrollments values(1,1),(2,2),(1,2);
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select student_name,course_name,faculty_name from students s join enrollments e on s.student_id=e.student_id join courses c on e.course_id=c.course_id join faculty f on c.faculty_id=f.faculty_id;
+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Anu          | DBMS        | Dr. Rao      |
| Bhanu        | OS          | Dr. Iyer     |
| Anu          | OS          | Dr. Iyer     |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> start transaction; insert into accounts values(104,'Anil',25000); commit;
Query OK, 0 rows affected (0.00 sec)

ERROR 1062 (23000): Duplicate entry '104' for key 'accounts.PRIMARY'
Query OK, 0 rows affected (0.00 sec)

mysql> start transaction; insert into accounts values(105,'Ravi',10000); rollback;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)

Query OK, 0 rows affected (0.05 sec)

mysql> start transaction; update accounts set balance=balance-5000 where account_no=101; update accounts set balance=balance+5000 where account_no=102; commit;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.05 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.01 sec)

mysql> start transaction; update accounts set balance=balance-3000 where account_no=101; update accounts set balance=balance+3000 where account_no=999; rollback;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> start transaction; update accounts set balance=balance-3000 where account_no=101; update accounts set balance=balance+3000 where account_no=999; rollback;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

Query OK, 0 rows affected (0.05 sec)

mysql> start transaction; update accounts set balance=balance+2000 where account_no=101; savepoint s1; update accounts set balance=balance-1000 where account_no=101; rollback to s1; commit;
Query OK, 0 rows affected (0.00 sec)

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.04 sec)

Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
mysql> mysql> select * from accounts;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'mysql> select * from accounts' at line 1
mysql> select * from accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101 | Ramesh | 27000.00 |
| 102 | Kavya | 20000.00 |
| 103 | Deepak | 20000.00 |
| 104 | Anil | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)
```